



WOLLO University

Department of software engineering

Exit multiple choice questions for

AI and big data

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1. a) Artificial intelligence is defined as a field aiming to make humans more intelligent.
2. b) Artificial intelligence is defined as a field aiming to improve security.
3. c) Artificial intelligence is defined as a field aiming to mine the data.
4. d) Artificial intelligence is defined as a field aiming to develop intelligent machines.

Answer with explanation: The correct answer is option d, artificial intelligence is defined as a field aiming to develop intelligent machines. It is basically the development of intelligent systems that can work and reach similar to human beings.

Q 3 Which is considered the branch of Artificial Intelligence?

1. a) Cyber Forensics
2. b) Machine Learning
3. c) Network Design
4. d) Full-stack Developer

Answer with explanation: The correct answer is option b, machine learning. It is the most important sub-area of Artificial Intelligence in which ML models are built or trained to do certain tasks.

Q. 4 Blind Search is used for which of the mentioned situations?

1. a) Advanced Game Theory
2. b) Real-life Simulation
3. c) Small Search Space
4. d) None of the above

Answer with explanation: The correct answer is option c, Small Search Space. Now that Blind Search does not contain domain information, it is best used for small Search Space.

Q. 5 In case a machine is capable of changing its course of action based on the external environment without any external help then the machine is called ____.

1. a) Intelligent
2. b) Mobile
3. c) Both A and B
4. d) None of the above

Answer with explanation: The correct answer is option a, Intelligent. When a machine is capable of changing its course of action without any external help the same is said to be an intelligent machine.

Q. 6 What is the main aim of Artificial Intelligence?

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1. a) To solve real-world issues
2. b) To explain different sorts of intelligence
3. c) To solve artificial problems
4. d) To obtain information about scientific causes

Answer with explanation: The correct answer is option b. the main aim of artificial intelligence is considered to explain different sorts of intelligence around the world.

Q. 7 Artificial Intelligence is classified into ____ category processes.

1. a) 2
2. b) 5
3. c) 3
4. d) 4

Answer with explanation: The correct answer is option c. Sensing, reasoning, and acting are three category processes into which Artificial Intelligence is classified.

Q. 8 ____ is the common language for AI.

1. a) Lisp
2. b) Python
3. c) PHP
4. d) Java

Answer with explanation: The correct answer is option b. While it is possible to do programming in any of the above-mentioned languages, Python has emerged as the go-to language for AI.

Q. 9 Which of the following are the proposition symbols in Artificial Intelligence?

1. a) true, false, and null
2. b) true
3. c) false
4. d) true and false

Answer with explanation: The correct answer is option d. True and false are the two proposition symbols in Artificial Intelligence.

Q. 10 ____ is not an application of AI.

1. a) Database Management System
2. b) Digital Assistants

3. c) Natural language processing
4. d) Computer Vision

Answer with explanation: The correct answer is option a. Except for Database Management System, all other mentioned options are applications of AI.

Q. 11 ____ is the informed search method.

1. a) Memory Bound Heuristic Search
2. b) A * Search
3. c) Best First Search
4. d) All of the above

Answer with explanation: The correct answer is option d. All the above-mentioned options are the informed search methods that are used in Artificial Intelligence.

Q. 12 ____ is considered a type of artificial intelligence agent.

1. a) Simple Reflex AI Agent
2. b) Learning AI agent
3. c) Goal-Based AI Agent
4. d) All of the above

Answer with explanation: The correct answer is option d. There are a total of 5 AI agents and these are Goal-Based AI Agent, Learning AI agent, Simple Reflex AI Agent, Utility-based Agent, and Model-based Reflex AI Agent.

Q. 13 ____ is a component of AI.

1. a) Training
2. b) Designing
3. c) Learning
4. d) Puzzling

Answer with explanation: The correct answer is option c. Learning is a process of gaining knowledge and enhancing awareness and thus, is an important component of AI.

Q 14 On which approach the face recognition system is based?

1. a) Weak AI Approach
2. b) Cognitive AI Approach
3. c) Strong AI Approach

4. d) Applied AI Approach

Answer with explanation: The correct answer is option d. the main aim of the Applied AI Approach is to produce commercially viable 'smart' systems.

Q. 15 The recognition system in AI is of how many types?

1. a) 3
2. b) 1
3. c) 5
4. d) 4

Answer with explanation: The correct answer is option a. the three types of recognition systems in AI are Content-based Image Retrieval, Biometric Identification, and Handwriting Recognition.

Q. 16 Procedural Domain Knowledge in a rule-based system is classified in the form of ____.

1. a) Control Rules
2. b) Production Rules
3. c) Meta Rules
4. d) None of the above

Answer with explanation: The correct answer is option b. Procedural Domain Knowledge in a rule-based system is classified in the form of Production Rules.

Q . 17 ____ was originally called 'the Imitation game'.

1. a) The Turning Test
2. b) LISP
3. c) The Halting Problem
4. d) None of the above

Answer with explanation: The correct answer is option a. The Turning Test was originally called 'the Imitation game'.

Q. 18 AI agent can interact with its environment by using ____.

1. a) only perceivers
2. b) only sensors
3. c) sensors and perceivers
4. d) None of the above

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Answer with explanation: The correct answer is option c. An AI agent can interact with its environment by using both sensors and perceivers. Sensors help it sense the surroundings and perceivers help it to act on them.

Q. 19 There are ____ types of observing environments?

1. a) 4
2. b) 3
3. c) 2
4. d) 0

Answer with explanation: The correct answer is option is c. there are two types of observing environments and these are Fully and Partial environments.

Q. 20 ____ are the compositions for Artificial Intelligence Agents.

1. a) Only Program
2. b) Only Architecture
3. c) Both Program and Architecture
4. d) None of the above

Answer with explanation: The correct answer is option c. Both Program and Architecture are the compositions for Artificial Intelligence Agents.

Q. 21 ____ is an example of an Artificial Intelligence Agent.

1. a) Human
2. b) Autonomous spacecraft
3. c) Robot
4. d) All of the above

Answer with explanation: The correct answer is option d. All the above-mentioned options are examples of Artificial Intelligence Agent. According to a report, humans can also be considered AI agents.

Q. 22 ____ is considered the expansion of Artificial intelligence application.s

1. a) Planning and Scheduling
2. b) Diagnosis
3. c) Game Playing
4. d) All of the above

Answer with explanation: The correct answer is option d. All the above-mentioned options considered the expansion of Artificial intelligence applications. In today's world, all sectors require intelligence and automation for better working.

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Q. 23 ____ environment is considered strategic.

1. a) Partial
2. b) Stochastic
3. c) Deterministic
4. d) Rational

Answer with explanation: The correct answer is option c. A deterministic environment is considered strategic because in this environment the output is determined based on a specific state.

Q. 24 Which of the following is the function of an AI Agent?

1. a) To map the percept sequence to an action
2. b) To map the goal sequence to an action
3. c) To work without direct interference from any external factor like humans
4. d) To map the environment sequence to an action

Answer with explanation: The correct answer is option a. The function of an AI agent is to map the percept sequence to an action.

1. The creator of _____ called it the “imitation game” initially.

- a. The Logic Theorist
- b. The Turing Test
- c. Cybernetics
- d. LISP

Answer: (b) The Turing Test

2. When someone programs a robot by moving it physically through the trajectory that they want it to follow is known as:

- a. robot vision control
- b. contact sensing control
- c. pick-and-place control

d. continuous-path control

Answer: (d) continuous-path control

3. The function that assigns the symbol a to b in LISP is:

a. (setq b 'a')

b. (setq b = 'a')

c. (setq b a)

d. (set b = 'a')

Answer: (a) (setq b 'a')

4. The computer program simulating the thought process of humans is known as:

a. Expert reason

b. Personal information

c. Expert system

d. Human logic

Answer: (c) Expert System

5. Natural language processing has two subfields, namely:

a. algorithmic and heuristic

b. time and motion

c. understanding and generation

d. symbolic and numeric

Answer: (c) understanding and generation

6. The bidirectional feedback loop helps link the computer modelling with:

a. cognitive science

b. human intelligence

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c. heuristic processing

d. artificial science

Answer: (a) cognitive science

7. The technique of computer vision that relies on the image templates is known as:

a. model-based vision

b. binocular vision

c. robot vision

d. edge detection

Answer: (a) model-based vision

8. Out of these schools, which one was NOT one of the early leaders in the field of AI research?

a. Stanford University

b. Massachusetts Institute of Technology

c. Harvard University

d. Dartmouth University

Answer: (c) Harvard University

9. For a speech understanding system to achieve widespread acceptance in the field of office automation, they must have:

a. isolated word recognition

b. speaker dependence

c. speaker independence

d. All of the above

Answer: (c) speaker independence

10. A blind search will be acceptable when this situation occurs:

- a. small search space
- b. complex game
- c. real-life situation
- d. All of the above

Answer: (a) small search space

11. The following function In LISP would return:

(minusp (-20 4 8 8 1))

- a. F
- b. T
- c. -20
- d. NIL

Answer: (b) T

12. Daniel Bobrow developed an AI system to read algebra word problems and solve them, known as:

- a. BACON
- b. SHRDLU
- c. STUDENT
- d. SIMD

Answer: (c) STUDENT

13. The elementary units of a language that are smaller than a word are known as:

- a. syllables
- b. phonemes
- c. allophones
- d. All of the above

Answer: (d) All of the above

14. The atom _____ stands for “true” in LISP.

- a. y
- b. time
- c. ml
- d. t

Answer: (d) t

15. What was the name of the very first AI programming language?

- a. LISP
- b. IPL
- c. FORTRAN
- d. BASIC

Answer: (b) IPL

16. _____ or modelling is the method using which we program a computer to exhibit human intelligence.

- a. Simulation
- b. Psychic Amelioration
- c. Duplication
- d. Cognitization

Answer: (a) Simulation

17. The Computer-Assisted Instruction (CAI) technique based on the programmed instruction is:

- a. problem-solving CAI
- b. generative CAI
- c. intelligent CAI

d. frame-based CAI

Answer: (d) frame-based CAI

18. The other name for a robot's "arm" is its:

a. Manipulator

b. End Effector

c. Servomechanism

d. Actuator

Answer: (a) Manipulator

19. KEE is a product of:

a. Tech knowledge

b. Texas Instruments

c. IntelliCorp

d. Teknowledge

Answer: (c) IntelliCorp

20. The _____ technique is currently investigated as the approach that helps in automatic programming.

a. specification by example

b. generative CAI

c. non-hierarchical planning

d. All of the above

Answer: (a) specification by example

21. A Newell and Simon program proved the theorems of Principia Mathematica. It was:

a. Logic Theorist

- b. Elementary Perceiver
- c. Boolean Algebra
- d. General Problem Solver

Answer: (a) Logic Theorist

22. We can characterise the IBM 3090, cray X-MP, and connection machine as:

- a. MISD
- b. SISD
- c. MIMD
- d. SIMD

Answer: (d) SIMD

23. We can categorise the BBN Butterfly, Cosmic Cube, Cedar, and Hypercube machine as:

- a. MISD
- b. SIMD
- c. MIMD
- d. SISD

Answer: (c) MIMD

24. The Automatic Reasoning Tool (ART) can be used on:

- a. microcomputers
- b. LISP machines
- c. personal computers
- d. All of the above

Answer: (b) LISP machines

25. One of the definitions of Artificial Intelligence focuses on various problem-solving methods used in processing:

- a. touch
- b. algorithms
- c. smell
- d. symbols

Answer: (d) symbols

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Data in ____ bytes size is called Big Data.

- A. Tera
- B. Giga
- C. Peta
- D. Meta

Answer: C) Peta

Explanation:

Big Data refers to data that is larger than a petabyte in size. The estimated volume of data that will be processed by Big Data solutions is significant and expected to continue to grow. In addition to increased storage and processing requirements, large data volumes necessitate the implementation of extra data preparation, curation, and management activities.

2. How many V's of Big Data?

- A. 2
- B. 3
- C. 4
- D. 5

Answer: D) 5

Explanation:

There are five V's of big data; these are Volume, Velocity, Variety, Value and Veracity. Knowing the 5 V's enables data scientists to extract more value from their data while also enabling the scientists' organizations to become more customer-centric as a result of their knowledge.

3. Unprocessed data or processed data are observations or measurements that can be expressed as text, numbers, or other types of media.

- A. True
- B. False

Answer: A) True

Explanation:

Unprocessed data or processed data are observations or measurements that can be expressed as text, numbers, or other types of media. In statistics, a data point, or observation, is a collection of one or more measurements taken on a single member of the observation unit (or unit of observation). Example: If the unit of observation is an individual and the research question is the determinants of money demand, a data point can be the values of income, wealth, the individual's age, and the number of dependents.

4. In computers, a ____ is a symbolic representation of facts or concepts from which information may be obtained with a reasonable degree of confidence.

- A. Data
- B. Knowledge
- C. Program
- D. Algorithm

Answer: A) Data

Explanation:

Information can be derived from data in computing if the data provides a symbolic representation of facts or concepts from which some probability can be calculated. While the summarizing of very large data sets might result in smaller data sets that are primarily composed of symbolic data, symbolic data are distinct in their own right on any sized data set, no matter how large or tiny it is.

5. In Big Data environments, Velocity refers –

- A. Data can arrive at fast speed
- B. Enormous datasets can accumulate within very short periods of time
- C. Velocity of data translates into the amount of time it takes for the data to be processed
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Large datasets can accumulate in a short amount of time in Big Data environments, since data arrives at lightning speed and accumulates in massive quantities. From the perspective of an enterprise, the velocity of data can be defined as the amount of time it takes for data to be processed once it enters the business's perimeter. In order to keep up with the rapid input of data, businesses must develop highly elastic and available data processing solutions, as well as the associated data storage capacities.

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6. In Big Data environments, Variety of data includes –

- A. Includes multiple formats and types of data
- B. Includes structured data in the form of financial transactions,
- C. Includes semi-structured data in the form of emails and unstructured data in the form of images
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

A wide range of data formats and kinds are required to be supported by Big Data systems, and this is referred to as data diversity. Enterprises have a number of issues when it comes to data integration, transformation, processing, and storage because of the variety of data. For example, financial transactions may contain structured data, while emails and photos may contain semi-structured data and unstructured data, respectively.

7. In Big Data environment, Veracity of data refers -

- A. Quality or fidelity of data
- B. Large size of the data that cannot be process
- C. Small size of the data that can easily process
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

The quality or fidelity of data is referred to as veracity. Data entering Big Data environments must be evaluated for quality, which may necessitate data processing activities in order to resolve erroneous data and remove noise from the data stream. When it comes to authenticity, data can be either part of the signal or part of the noise of a dataset.

8. Which of the following are Benefits of Big Data Processing?

- A. Cost Reduction
- B. Time Reductions
- C. Smarter Business Decisions
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Cost reduction, time reductions and smarter business decisions are the benefits of big data processing.

9. Structured data conforms to a data model or schema and is often stored in tabular form.

- A. True
- B. False

Answer: A) True

Explanation:

Structured data is data that has been organized according to a data model or schema and is frequently kept in tabular format. Due to the fact that it is used to record relationships between distinct things, it is most typically kept in a relational database. Enterprise applications and information systems, such as ERP and CRM systems, are frequently responsible for the generation of structured data.

10. Data that does not conform to a data model or data schema is known as _____.

- A. Structured data
- B. Unstructured data
- C. Semi-structured data
- D. All of the mentioned above

Answer: B) Unstructured data

Explanation:

It is referred to as unstructured data when the data does not comply to a data model or a data schema. Unstructured data is believed to account for 80 percent of all data within a given organization, according to some estimates. The growth rate of unstructured data is faster than that of structured data. SQL cannot be used to process or query unstructured data since it is not structured. As a binary large object, it is saved in a table in a relational database if it is necessary to be stored within the database (BLOB).

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11. Amongst which of the following is/are not Big Data Technologies?

- A. Apache Hadoop
- B. Apache Spark
- C. Apache Kafka
- D. Apache Pytarch

Answer: D) Apache Pytarch

Explanation:

Apache Pytarch is not a Big Data technology in the traditional sense. As part of a big data solution, Apache Hadoop, Apache Spark, and Apache Kafka are utilized. The Hadoop Distributed File System (HDFS) and a data processing engine that executes the MapReduce program to filter and sort data are the two primary components of Apache Hadoop. HDFS is a distributed file system that stores and distributes data across several computers. Apache Spark can also be used in conjunction with HDFS or another distributed file system. Hadoop MapReduce is capable of processing significantly larger data sets than Spark, particularly when the total size of the data collection exceeds the amount of memory that is available.

12. _____ involves the simultaneous execution of multiple sub-tasks that collectively comprise a larger task.

- A. Parallel data processing
- B. Single channel processing
- C. Multi data processing
- D. None of the mentioned above

Answer: A) Parallel data processing

Explanation:

Parallelism, which is defined in computing as the simultaneous execution of multiple processes, is the fundamental notion behind parallel data analysis. Parallel data processing is the simultaneous execution of several sub-tasks that together represent a bigger task in the context of a larger task. Parallel data analysis is a technique for analyzing data by running parallel processes on numerous computers at the same time.

13. Amongst which of the following can be considered as the main source of unstructured data.

- A. Twitter
- B. Facebook
- C. Webpages
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Unstructured data is primarily derived from social media platforms such as Twitter, Facebook, and the Internet. In the context of data storage, unstructured data refers to information that has not been organized according to a pre-determined data model or schema, and hence cannot be stored in a standard relational database management system (RDBMS). Text and multimedia are two types of unstructured content that are frequently encountered. Many business documents, as well as email messages, videos, images, webpages, and audio files, are unstructured in their content.

14. Amongst which of the following shows an example of unstructured data,

- A. Students roll number, age
- B. Videos
- C. Audio files
- D. Both B and C

Answer: D) Both B and C

Explanation:

Unstructured data includes files such as audio and video files, to name a few examples. In contrast to structured data, unstructured data does not fit well into a spreadsheet or database. It might be either textual or non-textual in nature. It can be created by a human or by a machine. Audio and video files, photos, text files - Word docs,

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PowerPoint presentations, email, chat logs, and other types of unstructured data are examples of unstructured data. Based on information gathered from social networking sites such as Facebook, Twitter, and LinkedIn Text messages, geolocation, chat, and call records are all examples of mobile data.

15. Scalability, elasticity, resource pooling, self-service, low cost and fault tolerance are the features of,

- A. Cloud computing
- B. Power BI
- C. System development
- D. None of the mentioned above

Answer: A) Cloud computing

Explanation:

Cloud computing is characterized by its scalability, elasticity, resource pooling, self-service, cheap cost, and fault tolerance, among other characteristics. While "Big Data" refers to massive volumes of data that have been collected, cloud computing refers to the technology that remotely receives this data and conducts any actions that have been specified on that information.

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16. Amongst which of the following is/are the cloud deployment models,

- A. Public Cloud
- B. Private Cloud
- C. Hybrid Cloud
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Public Cloud, Private Cloud and Hybrid Cloud are the cloud deployment models. A cloud deployment model is characterized by the location of the infrastructure that will be used for the deployment and the authority that will be exercised over that infrastructure.

17. Virtualization separates resources and services from the underlying physical delivery environment.

- A. True
- B. False

Answer: A) True

Explanation:

Virtualization is the process of isolating resources and services from the physical delivery environment that they are delivered in. Big data virtualization is a method that focuses on the creation of virtual structures for large-

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scale data storage and processing environments. The usage of big data virtualization can be beneficial to businesses and other organizations because it helps them to make use of all of the data assets they have collected in order to achieve a variety of goals and objectives.

18. What is a Virtual Machine (VM)?

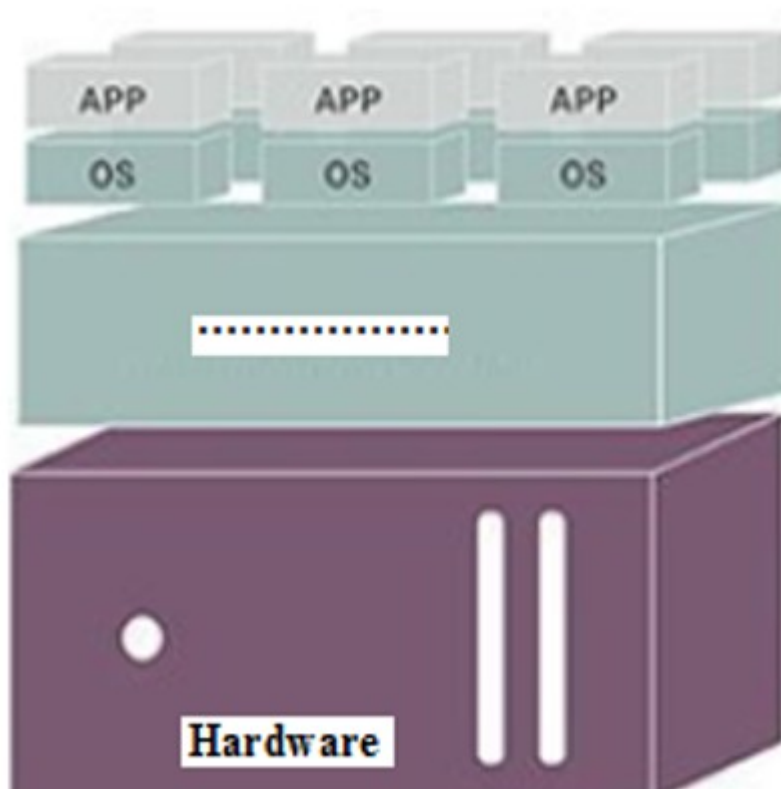
- A. Virtual representation of a physical computer
- B. Virtual representation of a logical computer
- C. Virtual System Integration
- D. All of the mentioned above

Answer: A) Virtual representation of a physical computer

Explanation:

A virtual machine (VM) is a representation of a physical computer that exists only in virtual space. In addition to a CPU, memory, and discs to store your stuff, it is capable of connecting to the internet if necessary. In contrast to the real and tangible components of your computer (referred to as hardware), virtual machines (VMs) are typically conceived of as virtual computers or software-defined computers that run on physical servers and exist solely as code.

19. In the given Virtual Architecture, name the missing layer,



- A. Virtualization layer
- B. Storage layer
- C. Abstract layer
- D. None of the mentioned above

Answer: A) Virtualization layer

Explanation:

It serves as an additional abstraction layer between network and storage hardware, processing, and the application executing on that hardware.... A machine that has a virtualization layer can create other (virtual) machines, which can then be used to install alternative operating systems on top of the main machine.

20. MongoDB is a ____ database.

- A. SQL
- B. DBMS
- C. NoSQL
- D. RDBMS

Answer: C) NoSQL

Explanation:

MongoDB is a NoSQL (non-relational) database. Databases that are not tabular in nature, such as NoSQL databases, store data in a different way than relational tables. NoSQL databases are classified into a number of categories based on the data model they use. Document, key-value, wide-column, and graph are the four most common types. They are capable of supporting variable schemas and scaling quickly when dealing with big amounts of data and significant user traffic.

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21. MongoDB support cross platform and is written in ____ language.

- A. Python
- B. C++
- C. R
- D. Java

Answer: B) C++

Explanation:

MongoDB is a cross-platform database that is created in the C++ programming language. MongoDB stores data as flexible, JSON-like documents, which means that fields can differ from document to document and that the data structure can be altered as the database matures. Because MongoDB is a distributed database at its heart, it comes with built-in features such as high availability, horizontal scaling, and geographic distribution that are simple to utilize.

22. Amongst which of the following is / are true to run MongoDB?

- A. High availability through built-in replication and failover
- B. Management tooling for automation, monitoring, and backup
- C. Fully elastic database as a service with built-in best practices
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

It is necessary to have high availability with built-in replication and failover as well as management tooling for automation, monitoring, and backup. It is also necessary to have MongoDB running as a fully elastic database as a service with built-in best practices.

23. Big data deals with high-volume, high-velocity and high-variety information assets,

- A. True
- B. False

Answer: A) True

Explanation:

Big data is defined as information assets with a high volume, a high velocity, and a high variety of information assets that necessitate the use of cost-effective, creative types of information processing in order to get greater insight and make better decisions.

24. _____ hypervisor runs directly on the underlying host system. It is also known as "Native Hypervisor" or "Bare metal hypervisor".

- A. TYPE-1 Hypervisor
- B. TYPE- 2 Hypervisor
- C. Both A and B
- D. None of the mentioned above

Answer: A) TYPE-1 Hypervisor

Explanation:

TYPE-1 The hypervisor is a virtual machine that runs on top of the underlying host system. It is also referred to as a "Native Hypervisor" or a "Bare metal hypervisor" in some circles.

25. _____ is also known as "Hosted Hypervisor".

- A. TYPE-1 Hypervisor
- B. TYPE- 2 Hypervisor
- C. Both A and B

D. None of the mentioned above

Answer: B) TYPE- 2 Hypervisor

Explanation:

The term "Hosted Hypervisor" refers to a TYPE-2 hypervisor that is hosted on a third-party server. Type 2 hypervisors are often encountered in setups with a modest number of servers, as the name suggests. The fact that we do not have to install a management console on another machine in order to set up and administer virtual machines is what makes them so convenient. All of this can be accomplished on the server where the hypervisor is installed. Hosted hypervisors are mostly used as management consoles for virtual machines, and we may do any activity with the help of the built-in functions of the hypervisor.

26. In the layered architecture of Big Data Stack, Interfaces and feeds,

- A. Internally managed data
- B. Data feeds from external sources.
- C. It provides access to each and every layer & components of big data stack
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Interfaces and feeds internally maintained data; data feeds from external sources; and offers access to each and every layer and component of the Big Data Stack are all included in the tiered design of the Big Data Stack. In order to create the Data Pipeline in accordance with the diverse requirements of either the Batch Processing System or the Stream Processing System, the Big Data Architecture is used. This architecture is made up of six layers, each of which is responsible for ensuring the secure transit of data.

27. _____ is the supporting physical infrastructure is fundamental to the operation and scalability of big data architecture.

- A. Redundant physical infrastructure
- B. Integrated System
- C. Integrated Database
- D. All of the mentioned above

Answer: A) Redundant physical infrastructure

Explanation:

It is critical for the functioning and scalability of big data architecture to have redundant physical infrastructure as a backbone of the infrastructure. Ideally, networks should be redundant and have sufficient capacity to handle the projected amount and velocity of inbound and outbound data, in addition to the "regular" network traffic encountered by the organization.

28. The physical infrastructure of a big data is based on a distributed computing model.

- A. True
- B. False

Answer: A) True

Explanation:

It is a distributed computing model that underpins the physical infrastructure of a big data system. The physical infrastructure will "make or break" the deployment of big data since it is concerned with high-velocity, high-volume, and high-variety data streams. The majority of big data deployments require high availability, which necessitates the use of resilient and redundant networks, servers, and physical storage systems. The concepts of resilience and redundancy are intertwined.

29. Security infrastructure refers the data about your constituents needs to be protected to ____.

- A. Meet compliance requirements
- B. Protect the privacy
- C. Both A and B
- D. None of the mentioned above

Answer: C) Both A and B

Explanation:

Security infrastructure refers to the data about your constituents that needs to be protected in order to comply with regulatory obligations and protect their personal information.

30. Reporting and visualization enables.

- A. Processing of data
- B. User friendly representation
- C. Both A and B
- D. None of the mentioned above

Answer: C) Both A and B

Explanation:

Reporting and visualization make it possible to process data and present it in a user-friendly manner. Putting data into a chart, graph, or other visual format that may be used to inform analysis and interpretation is known as data visualization (or data presentation). Different stakeholders can engage with and learn from data visualizations since they make the examined data easily understandable.

| ADVERTISEMENT

31. Data interpretation refers -

- A. Process of attaching meaning to the data

- B. Convert text into insightful information
- C. Effective conclusion
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Data interpretation is the process of giving meaning to information gathered from various sources. Interpretation entails drawing inferences about generalization, correlation, and causation, and it is designed to provide answers to critical learning questions regarding our research endeavor.

32. The significance of metadata is to provide information about a dataset's characteristics and structure.

- A. True
- B. False

Answer: A) True

Explanation:

It is possible to obtain information about the qualities and structure of a dataset by using metadata. This sort of data is usually generated by machines and can be added to other types of data. The tracking of metadata is critical to Big Data processing, storage, and analysis because it offers information about the data's origins and provenance during processing, which is otherwise impossible to obtain.

33. Data throttling refers to the performance of a solution is throttled,

- A. True
- B. False

Answer: A) True

Explanation:

There are answers to the business difficulty, however on traditional technology, the performance of a solution is throttled as a result of data accessibility, data latency, data availability, or bandwidth limitations in relation to the size of the inputs.

34. The Big data analytics work on the unstructured data, where no specific pattern of the data is defined.

- A. True
- B. False

Answer: A) True

Explanation:

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Big data analytics is used to analyze unstructured data, which is data that does not follow a predefined pattern and cannot be predicted. In this case, the information is not structured in rows and columns. The real-time flow of data is gathered, and analysis is performed on the data. When the amount of data to be analyzed is huge, efficiency increases.

35. Amongst which of the following represents the Use of Hadoop,

- A. Robust and Scalable
- B. Affordable and Cost Effective
- C. Adaptive and Flexible
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

There are numerous advantages to implementing Hadoop. The following are some of the major advantages: robustness and scalability, affordability and cost effectiveness, adaptability and flexibility, as well as high availability and fault tolerance. In order to operate Hadoop, we do not require any specialized hardware. We can just utilize a commodity server for this purpose.

1. Data in _____ bytes size is called Big Data.

- A. Tera
- B. Giga
- C. Peta
- D. Meta

View Answer

Ans : C

Explanation: data in Peta bytes i.e. 10^{15} byte size is called Big Data.

2. How many V's of Big Data

- A. 2
- B. 3
- C. 4
- D. 5

View Answer

Ans : D

Explanation: Big Data was defined by the “3Vs” but now there are “5Vs” of Big Data which are Volume, Velocity, Variety, Veracity, Value

3. Transaction data of the bank is?

- A. structured data
- B. unstructured datat
- C. Both A and B
- D. None of the above

View Answer

Ans : A

Explanation: Data which can be saved in tables are structured data like the transaction data of the bank.

4. In how many forms BigData could be found?

- A. 2
- B. 3
- C. 4
- D. 5

View Answer

Ans : B

Explanation: BigData could be found in three forms: Structured, Unstructured and Semi-structured.

5. Which of the following are Benefits of Big Data Processing?

- A. Businesses can utilize outside intelligence while taking decisions
- B. Improved customer service
- C. Better operational efficiency
- D. All of the above

View Answer

Ans : D

Explanation: All of the above are Benefits of Big Data Processing.

6. Which of the following are incorrect Big Data Technologies?

- A. Apache Hadoop
- B. Apache Spark
- C. Apache Kafka
- D. Apache Pytarch

View Answer

Ans : D

Explanation: Apache Pytarch is incorrect Big Data Technologies.

7. The overall percentage of the world's total data has been created just within the past two years is ?

- A. 80%
- B. 85%
- C. 90%
- D. 95%

View Answer

Ans : C

Explanation: The overall percentage of the world's total data has been created just within the past two years is 90%.

8. Apache Kafka is an open-source platform that was created by?

- A. LinkedIn
- B. Facebook
- C. Google
- D. IBM

View Answer

Ans : A

Explanation: Apache Kafka is an open-source platform that was created by LinkedIn in the year 2011.

9. What was Hadoop named after?

- A. Creator Doug Cutting's favorite circus act
- B. Cuttings high school rock band
- C. The toy elephant of Cutting's son
- D. A sound Cutting's laptop made during Hadoop development

View Answer

Ans : C

Explanation: Doug Cutting, Hadoop creator, named the framework after his child's stuffed toy elephant.

10. What are the main components of Big Data?

- A. MapReduce
- B. HDFS
- C. YARN
- D. All of the above

View Answer

Ans : D

Explanation: All of the above are the main components of Big Data.

11. All of the following accurately describe Hadoop, EXCEPT _____

- A. Open-source
- B. Real-time
- C. Java-based
- D. Distributed computing approach

View Answer

Ans : B

Explanation: Apache Hadoop is an open-source software framework for distributed storage and distributed processing of Big Data on clusters of commodity hardware.

12. _____ has the world's largest Hadoop cluster.

- A. Apple
- B. Datamatics
- C. Facebook
- D. None of the above

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View Answer

Ans : C

Explanation: Facebook has many Hadoop clusters, the largest among them is the one that is used for Data warehousing.

13. Facebook Tackles Big Data With _____ based on Hadoop.

- A. Project Prism
- B. Prism
- C. Project Big
- D. Project Data

View Answer

Ans : A

Explanation: Prism automatically replicates and moves data wherever it's needed across a vast network of computing facilities.

14. _____ is general-purpose computing model and runtime system for distributed data analytics.

- A. Mapreduce
- B. Drill
- C. Oozie
- D. None of the above

View Answer

Ans : A

Explanation: Mapreduce is general-purpose computing model and runtime system for distributed data analytics.

15. The examination of large amounts of data to see what patterns or other useful information can be found is known as

- A. Data examination
- B. Information analysis
- C. Big data analytics
- D. Data analysis

View Answer

Ans : C

Explanation: The examination of large amounts of data to see what patterns or other useful information can be found is known as Big data analytics.

16. Big data analysis does the following except?

- A. Collects data
- B. Spreads data
- C. Organizes data
- D. Analyzes data

View Answer

Ans : D

Explanation: Big data analysis does the following except Spreads data.

17. What makes Big Data analysis difficult to optimize?

- A. Big Data is not difficult to optimize
- B. Both data and cost effective ways to mine data to make business sense out of it
- C. The technology to mine data
- D. None of the above

View Answer

Ans : B

Explanation: Both data and cost effective ways to mine data to make business sense out of it makes Big Data analysis difficult to optimize.

18. The new source of big data that will trigger a Big Data revolution in the years to come is?

- A. Business transactions
- B. Social media
- C. Transactional data and sensor data
- D. RDBMS

View Answer

Ans : C

Explanation: The new source of big data that will trigger a Big Data revolution in the years to come is Transactional data and sensor data.

19. The unit of data that flows through a Flume agent is

- A. Log
- B. Row
- C. Record
- D. Event

View Answer

Ans : D

Explanation: The unit of data that flows through a Flume agent is Event.

20. Listed below are the three steps that are followed to deploy a Big Data Solution except

- A. Data Processing
- B. Data dissemination
- C. Data Storage
- D. Data Ingestion

View Answer

Ans : B

Explanation: Listed below are the three steps that are followed to deploy a Big Data Solution except Data dissemination.

1. Who was written Hadoop?

A. Java (software platform)

- B. Perl
- C. Lua (programming language)
- D. Java (programming language)
- E. All of these
- F. both A and C

Answer - Click Here:

D

2. Hadoop run Which of the following platforms?

- A. Cross-platform
- B. Debian
- C. Bare-metal
- D. Unix-like
- E. All of these
- F. both A and C

Answer - Click Here:

A

3. Reliability by replicating data achieves Hadoop the across multiple hosts, and then Hadoop not require _____ storage on hosts.

- A. Operating system
- B. Standard RAID levels
- C. ZFS
- D. RAID
- E. All of these
- F. both A and C

[su_spoiler title="Answer – Click Here:" class="padding: -10px;"]D[/su_spoiler]

4.

The HBase database includes the Hadoop list, the Apache Mahout _____ system, and matrix operations.

- A. Statistical classification
- B. Pattern recognition
- C. Machine learning
- D. Artificial intelligence
- E. All of these
- F. both D and C

Answer - Click Here:

C

5. After the Hadoop named?

- A. A sound Cutting's laptop made during Hadoop development
- B. Cutting's high school rock band
- C. The toy elephant of Cutting's son
- D. Creator Doug Cutting's favorite circus act

E. All of these

F.both Dand C

Answer - Click Here:

D

6. These are the given are completely describe Hadoop, EXCEPT:

A. Open-source

B. Real-time

C. Distributed computing approach

D. Java-based

E.All of these

F. both Dand C

Answer - Click Here:

B

7. The world's largest Hadoop cluster.

A. Apple

B. Facebook

C. Datamatics

D. None of the mentioned

E. All of these

F. both Dand C

Answer - Click Here:

B

8. The Big Data Tackles Facebook are based on on_____ Hadoop.

A. 'Project Data

B. 'Prism'

C. 'Project Big'

D. 'Project Prism'

E. All of these

F. both Dand C

Answer - Click Here:

D

9. The total size driven by the number of maps is usually of :

A. Inputs

B. asks

C. Outputs

D. None of the mentioned

E. All of these

F. both Dand C

Answer - Click Here:

A

10. The default Partitioner for partitioning keyspace Which of the following is correct.

A. HashPartitioner

- B. Partitioner
- C. HashPar
- D. None of the mentioned
- E. All of these
- F. both Dand C

Answer - Click Here:

A

11The mappers Input to the _____ are sorted by

- A. Shuffle
- B. Reducer
- C. Mapper
- D. All of the mentioned
- E. None of these
- F. both Dand C

Answer - Click Here:

B

12.The right number of reduces to be :

- A. 0.90
- B. 0.95
- C. 0.36
- D. 0.80
- E. All of these
- F. both Dand C

Answer - Click Here:

B

13. The partitioning of the keys is controlled by the intermediate map-outputs.

- A. Collector
- B. InputFormat
- C. Partitioner
- D .None of the mentioned
- E. All of these
- F. both Dand C

Answer - Click Here:

C

14.Which SequenceFile are present in Hadoop I/O ?

- A. 2
- B. 8
- C. 9
- D. 3
- E. All of these
- F. both Dand C

Answer - Click Here:

C

15. The two SequenceFile is directory contains that

- A. MapFile
- B. MapperFile
- C. ReduceFile
- D. None of the mentioned

E. All of these

F. both Dand C

Answer - Click Here:

A

16. Snappy compression is similar to?

- A. LZO
- B. Bzip2
- C. Gzip
- D. All of the mentioned

E. All of these

F. both Dand C

Answer - Click Here:

A

17. Splittable compression is supported Which of the following?

- A. Bzip2
- B. All of the mentioned
- C. Gzip
- D. LZO

E. both Dand C

F. All of these

Answer - Click Here:

D

18. slowest compression technique is _____?

- A. LZO
- B. Gzip
- C. Bzip2
- D. All of the mentioned

E. All of these

F. both Dand C

Answer - Click Here:

C

19. Gzip contain compressed files which is form _____ extension.

- A. gzp
- B. g
- C. gzip
- D. gz

E. both Dand C

F. All of these

Answer - Click Here:

D

20. DEFLATE algorithm based on the?

- A. Bzip2
- B. LZO
- C. Gzip
- D. All of the mentioned
- E. None of these
- F. both Dand C

Answer - Click Here:

C

21. Which of the following is a typically compresses files which are best available techniques.10% to 15 %.

- A. Bzip2
- B. LZO
- C. Gzip
- D. both Dand C
- E. All of the mentioned
- F. None of these

Answer - Click Here:

A

22. Which of the following is provides search technology? and Java-based indexing

- A. Solr
- B. Lucy
- C. Lucene Core
- D. None of these
- E. both Dand C
- F. All of the mentioned

Answer - Click Here:

C

23. Lucene Core is a high performance search server built using_____.

- A. PyLucene
- B. Lucy
- C. Lucene Core
- D. Solr
- E. None of these
- F. both Dand C

Answer - Click Here:

D

24. Subproject collecting and distributing free materials with the aim of

- A. ORP

- B. ORS
- C. OSR
- D. OPR
- E. None of these
- F. both Dand C
- G. All of the mentioned

Answer - Click Here:

A

25. the Core project is a Python port is _____.

- A. Lucene Core
- B. Solr
- C. PyLucene
- D. Lucy
- E. None of these
- F. both Dand C
- G. All of the mentioned

Answer - Click Here:

C

26. The size of text indexed of Lucene index size is roughly _____.

- A. 20%
- B. 10%
- C. 70%
- D. 50%
- E. None of these
- F. both Dand C
- G. All of the mentioned

Answer - Click Here:

A

27. To set the transaction isolation level which of the following is used?

- A. -truncateTable=[true/false]
- B. force=[true/false].
- C. -isolation=LEVEL
- D. -incremental=[true/false].
- E. None of these
- F. both Dand C
- G. All of the mentioned

Answer - Click Here:

C

28. Avro data as Hive tables to read or write _____ allows to users

- A. None of the mentioned
- B. SqlSerde
- C. HiveSerde
- D. AvroSerde
- E. None of these

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F. both D and C

Answer - Click Here:

D

29. The Hive table schema Starting in Hive the Avro schema can be inferred from the_____.

A. 0.11

B. 0.12

C. 0.14

D. 0.13

E. None of these

F. both D and B

Answer - Click Here:

C

30. Data type is supported by Hive which of the following is used _____ ?

A. string

B. None of these

C. map

D. enum

E. record

F. both D and B

Answer - Click Here:

D

31. How we converted data type is to Array prior to Hive 0.12.0?

A. map

B. long

C. bytes

D. float long

E. None of these

F. both D and B

Answer - Click Here:

C

32. Are defined with Avro schemas _____

A. JAVA

B. XML

C. All of the mentioned

D. JSON

E. None of these

F. both D and B

Answer - Click Here:

D

33. _____ of the field is used to Thrift resolves possible conflicts.

A. Name

B. UID

C. Static number

D. All of the mentioned

E. None of these

F. both D and B

Answer - Click Here:

C

34. _____ layer of is said to be the future Hadoop.Avro.

A.RMC

B. RPC

C. RDC

D. All of the mentioned

E. None of these

F. both D and B

Answer - Click Here:

B

35. Which of the following is used to declare the schema of our data either in a _____ file?

A. SQL

B. R

C. JSON

D. XML

E. None of these

F. both D and B

Answer - Click Here:

A

36. In _____ stage the framework groups Reducer inputs by key.

A. reduce

B. shuffle

C. sort

D. none of the mentioned

E.All of these

F.both D and B

Answer - Click Here:

C

37. High storage density Which of the following has high storage density?

A. RAM_DISK

B. ARCHIVE

C. ROM_DISK

D. All of the mentioned

E. None of these

F. both D and B

Answer - Click Here:

B

Which of the following is a wrong statement.

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- (A). The big volume actually represents Big Data
- (B). Big Data is just about tons of data
- (C). The data growth and social media explosion have improved that how we look at the data
- (D). All of these
- (E). None of these

Answer: b

Which of the following is a correct statement.

- (A). Machine learning emphasizes on prediction, based on well-known properties learned from the training data
- (B). Data Cleaning emphasizes on prediction, based on well-known properties learned from the training data
- (C). Both a and b
- (D). None of these

Answer: d

Beyond Volume, velocity and variety are the concerns and problems of big data veracity.

- (A). True
- (B). False
- (C). Partially True

Answer: a

Which of the big data characteristic is comparatively more concerned with data science?

- (A). Variety
- (B). Velocity
- (C). Volume
- (D). None of these

Answer: a

Which of the following analytical competencies are offered by information management corporations?

- (A). Stream Computing
- (B). Content Management
- (C). Information Integration
- (D). All of these

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(E). None of these

Answer: d

3V's are not enough to describe big data.

(A). True

(B). False

(C). Partially True

Answer: a

Which step is executed by the data scientist after obtaining the data?

(A). Data Replication

(B). Data Integration

(C). Data Cleansing

(D). All of these

(E). None of these

Answer: c

Which of the following emphasizes on the discovery of earlier properties that are not known on the data?

(A). Machine Learning

(B). Big Data

(C). Data wrangling

(D). Data mining

(E). None of these

Answer: d

Principal components or factor analytic models on covariates are often helpful for reducing the complex covariate spaces.

(A) . True

(B) . False

MCQ Answer: a

What is the total number of outcomes that are possible with the Bernoulli trial?

(A) . 1

(B) . 2

(C) . 3
(D) . None of these
MCQ Answer: b

Which of the following is a statistical process of analysis for guessing the relationships among variables?

(A) . Causal
(B) . Multivariate
(C) . Regression
(D) . All of these
(E) . None of these
MCQ Answer: c

Which of the following is the wrong statement.

(A) . The addition of squared terms makes it twice continuously differentiable at the knot points
(B) . The addition of squared terms makes it continuously differentiable at the knot points
(C) . typically, Asymptotics is used for inference
(D) . None of these
MCQ Answer: a

How many total components present in generalized linear models are?

(A) . 2
(B) . 4
(C) . 6
(D) . None of these
MCQ Answer: d

Which of the following is the wrong statement.

(A) . Transformations are often easy to interpret in a linear model
(B) . Additive response models don't make much sense if the response is discrete, or strictly positive
(C) . Regression models are helpful to predict one variable from one or more other variables
(D) . All of these
(E) . None of these
MCQ Answer: a

Which component is involved in generalized linear models?

(A) . An exponential family model for the response
(B) . A link function that connects the means of the response to the linear predictor
(C) . A systematic component via a linear predictor
(D) . All of these
(E) . None of these
MCQ Answer: d

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Collection of exchangeable binary outcomes for the same covariate data are commonly known as which of the following outcomes.

- (A) . random
- (B) . binomial
- (C) . direct
- (D) . None of these

MCQ Answer: b

Linear models are very helpful models for applied statistical techniques.

- (A) . True
- (B) . False

MCQ Answer: b

Which is an example use of Poisson distribution?

- (A) . Analyzing contingency table data
- (B) . Incidence rates
- (C) . Modeling web traffic hits
- (D) . All of these

MCQ Answer: d

Which of the following is the wrong statement.

- (A). Causal relationships may not relate to every individual
- (B). Complication approached exist for inferring causation
- (C). Randomized studies are not used to identify causation
- (D). All of these

MCQ Answer: c

Which data mining technique is used to uncover patterns in the given data?

- (A). Data Dredging
- (B). Data booting
- (C). Data merging
- (D). Data bagging

MCQ Answer: a

If A predicts B, it does mean A causes B.

- (A). True
- (B). False

MCQ Answer: b

Which approach should be used if we can't fix the variable?

- (A). non stratify the variable
- (B). randomize the variable
- (C). generalize the variable
- (D). None of these

MCQ Answer: b

Which is an ideal methodology for performing experiments in data science?

- (A). Measure variability
- (B). Generalize the problem
- (C). Have Replication
- (D). All of these

MCQ Answer: d

Which of the following is the correct statement.

- (A). If equations and parameter are not, they may be concluded with data analysis
- (B). If equations are not known but the parameters are, they may be concluded with data analysis
- (C). If equations are known but the parameters are not, they may be concluded with data analysis
- (D). None of these

MCQ Answer: c

Which is the top most significant entity in data science?

- (A). question
- (B). answer
- (C). data
- (D). None of these

MCQ Answer: a

Which is commonly called 'data fishing'?

- (A). Data bagging
- (B). Data booting

(C). Data merging

(D). None of these

MCQ Answer: d